



United International University (UIU)
Department of Computer Science and Engineering
CSE 1325: DIGITAL LOGIC DESIGN, Final Fall 2024

Total Marks: **40** Duration: 2 hours

Answer ALL Questions

**Any examinee found adopting unfair means will be expelled from the trimester /
program as per UIU disciplinary rules**

1. Imagine you are designing a circuit for a security system in a futuristic vault. The vault requires a secret key to unlock it, but the key is encoded as a sequence of bits that is transmitted one bit at a time. The security system unlocks the vault if and only if it detects the sequence "1011" in the incoming bit stream. If the sequence is found, the system sends a signal of 1 to the output to unlock the vault; otherwise, it continuously outputs 0. Do the following.
 - A. Design the state diagram for the sequence recognizer. [2]
 - B. Create a state table for the circuit. [2]
 - C. Derive the minimized logic equations for the flip-flops inputs and output. [3]
 - D. Draw the circuit diagram for the sequence recognizer. [1]

2. In the bustling streets of Dhaka, the Bangladesh Traffic Control Authority (BTCA) is working on a smart traffic management system to ease congestion at major intersections like Gulshan Circle. They plan to use a digital counter to manage traffic light sequences. The counter must follow this specific sequence to optimize traffic flow: $1 \rightarrow 4 \rightarrow 6 \rightarrow 7 \rightarrow 5 \rightarrow 2 \rightarrow 3 \rightarrow 1$ (and repeat). The BTCA has asked a group of students from UIU to design the counter. Their mentor, Professor XX, instructed them to use T flip-flops to design and ensure the counter is synchronous and cost-effective for real-world deployment.

Tasks:

 - A. Analyze the sequence and draw the state table. [2]
 - B. Derive the Flip-Flop excitation table. [2]
 - C. Simplify using K-Maps. [3]
 - D. Draw the logic diagram. [1]

3.
 - A. You need to design an 8x1 MUX but you have 2x1 MUX only. Considering that you have got an infinite supply of 2x1 MUX, with proper steps show how you can implement it. [4]
 - B. Implement the following two functions in a single circuit using 3-to-8 decoders and basic logic gates: [4]
 - i. $F1(A, B, C, D) = A'B'C'D + A'BCD + AB'C'D + AB'CD$
 - ii. $F2(A, B, C, D) = \sum_m(0, 2, 14, 15)$

4. An earthquake detection system is being implemented for six major cities in Bangladesh. Each city has a connected earthquake sensor, which triggers the encoder when an earthquake is detected. The priority of detection is ordered as follows:

Dhaka > Chittagong > Khulna > Sylhet > Rajshahi > Barishal

When multiple sensors are active simultaneously, the system should prioritize the city with the highest importance. Additionally, the output should include an indicator that becomes active if any of the sensors are triggered; otherwise, it remains inactive.

Cities are marked using the first alphabet of their name and are associated with a unique 3-bit code as follows: D - 110, C - 100, K - 000, S - 010, R - 111, B - 101.

You have to answer the followings:

- A. Derive the truth table of the circuit including the indicator bit. [3]
 - B. Derive the Boolean expressions for all the outputs. [3]
 - C. Draw the logic diagram using basic gates. [2]
5. Design a 3-bit universal shift register using D flip-flops. The register should have two control lines, S_1 and S_0 , which determine the operation of the register. The functionality of the register based on the control lines is as follows: [8]

S_1	S_0	Operation
1	1	Shift the contents of the register left by one position and insert 1 into the least significant bit (LSB)
1	0	Clear all flip-flops in the register to 0
0	1	Perform parallel loading.
0	0	No change

Excitation Tables for Flip-Flops

$Q(t)$	$Q(t+1)$	T	Operation
0	0	0	No change
0	1	1	Complement
1	0	1	Complement
1	1	0	No change

$Q(t)$	$Q(t+1)$	D	Operation
0	0	0	Reset
0	1	1	Set
1	0	0	Reset
1	1	1	Set

$Q(t)$	$Q(t+1)$	J	K	Operation
0	0	0	x	No change/reset
0	1	1	x	Set/complement
1	0	x	1	Reset/complement
1	1	x	0	No change/set